

An Efficient Approach to Detecting Lung Nodules Using Swin Transformer

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Introduction

- The most common cancer death cause is **lung cancer** (20.4% of cancer deaths) [1].
- Lung nodules** are small abnormal growths in the lungs that can be either cancerous or non-cancerous [1].
- Early-stage detection of lung nodules is **significantly crucial** for increasing survival rate [3].
- If nodules aren't **diagnosed promptly**, they will turn into lumps which bring complicated treatments [2].
- Nodules appear in different sizes, especially small ones make detection more difficult [1].
- Automatic lung nodule detection could bring ease for radiologists to identify cancerous nodules [4].
- Deep Learning methods have better feature extraction & detection ability compared to traditional methods.

Background

- All previous works utilized the 3D structure of the scans, and consequently, 3D object detection models, which results in high computational and time costs.
- No examination of model performance in detecting small nodules.
- LUNA16 dataset: 888 scans containing 1186 nodules with 512×512 slices.

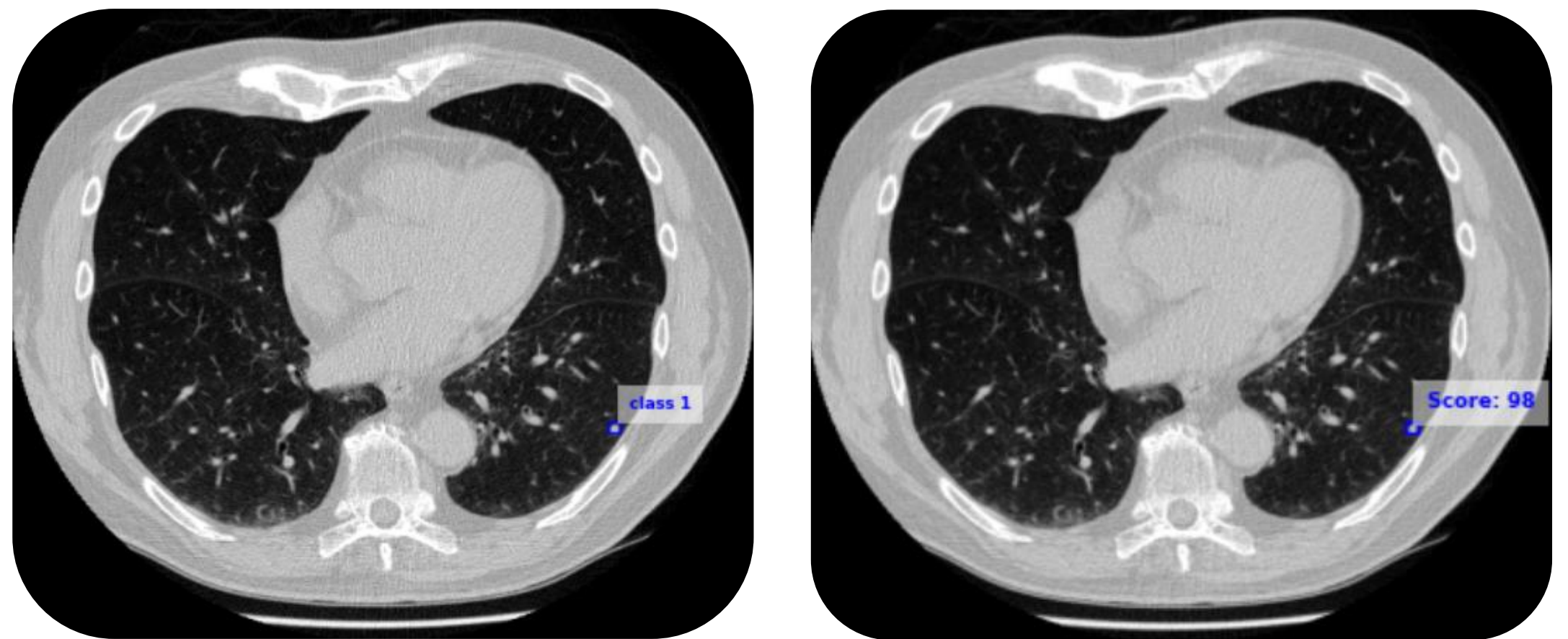


Fig. 1: 2D slices of a CT scan. Left: nodule in dataset, right: detected nodule.

Proposed model

- Extracting 2D slices from the 3D CT scans.
- Utilizing 2D models as opposed to 3D models used previously.
 - ✓ Significantly reducing training and inference costs.
- Employing Swin Transformer and Feature Pyramid Network.
 - ✓ Using features of all layers for better detection of small nodules.
 - ✓ Leveraging the benefits of multi-head self-attention.
 - ✓ Lowering computational costs, unlike Vision Transformers.
- Using a Transfer Learning approach in training
 - ✓ 5 epochs, as opposed to 100 to 300 in previous works.

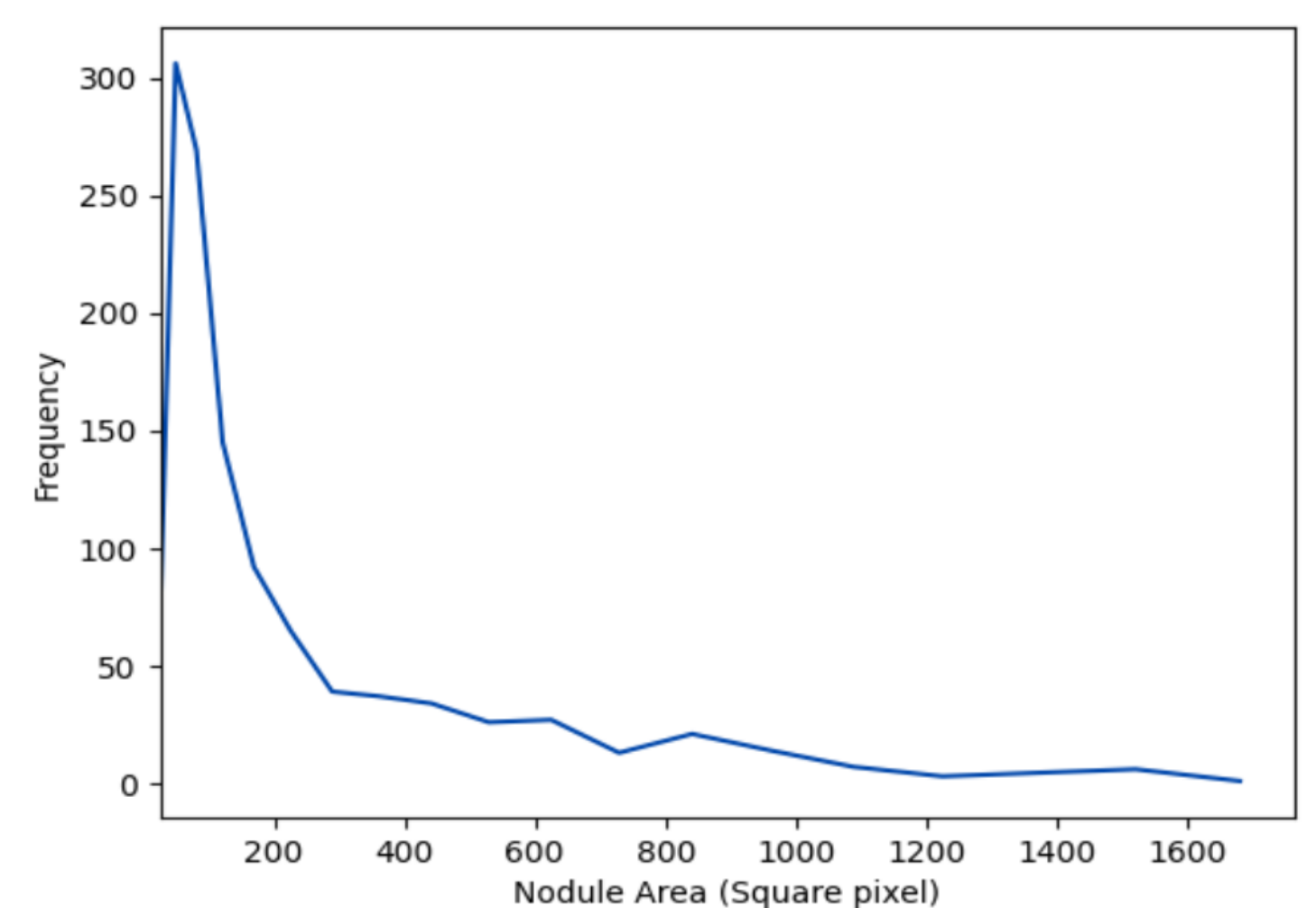


Fig. 2: Distribution of nodule areas.

Table. 1: Comparison of our model with others under different IoU thresholds OF 0.5 to 0.95.

Model	Area	mAP	mAR
[5]	Small	0.704	0.706
	All	0.797	0.803
[6]	Small	0.719	0.720
	All	0.807	0.810
[7]	Small	0.899	0.901
	All	0.944	0.947
Ours	Small	0.912	0.917
	All	0.947	0.949

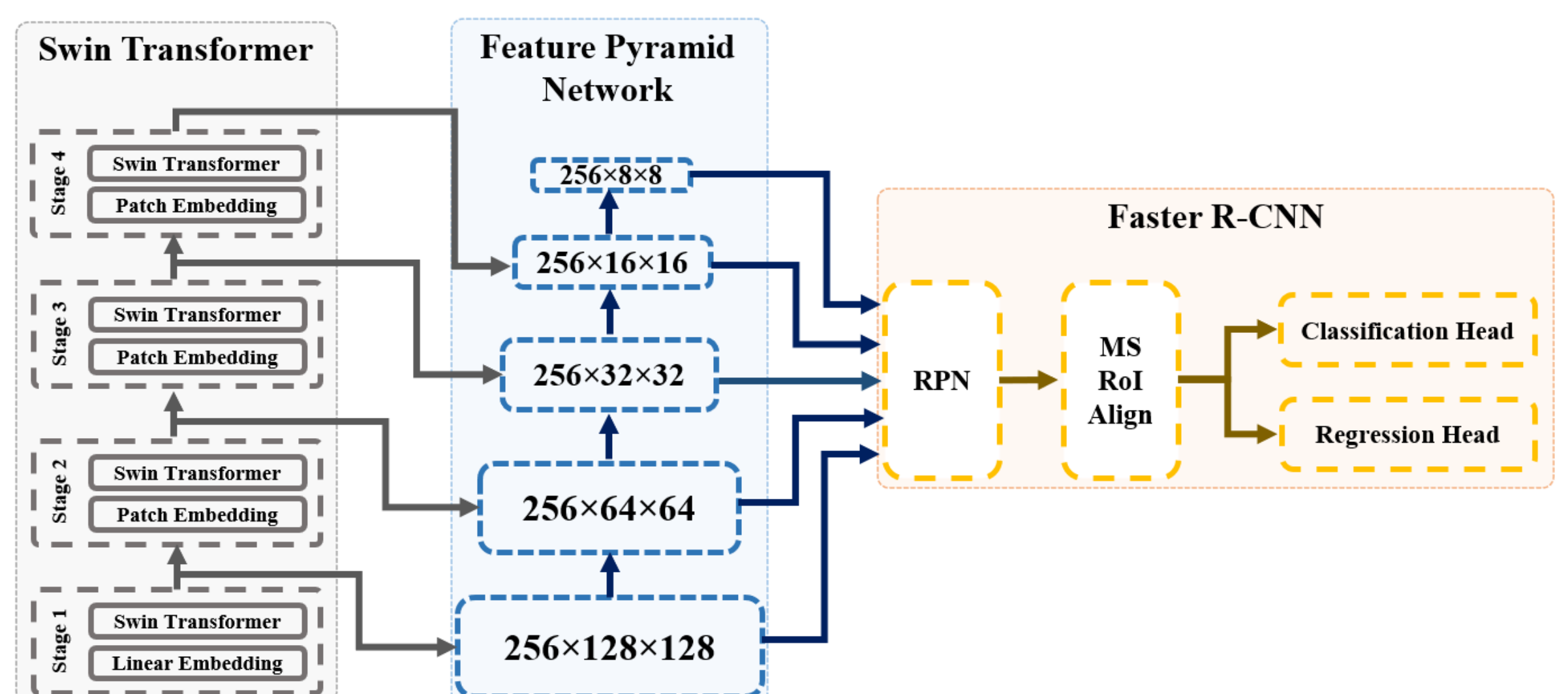


Fig. 3: The overall structure of the proposed model.

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